Approach to Skin Lesions

Dr. Taha Habibullah, Consultant Dermatology | Source: `Approach_to_Skin_Lesions.pdf` (56 slides)

> **Legend:** ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The lecture has **no answer key** for its case; the answer below is **[reasoning added; key not on slides]**.
>
> 🔗 **Bridge from earlier lectures:**
>
> - **Pituitary 1 / Adrenal 1 / Adrenal 2: hirsutism** work-up by source:
> - ovary: **PCOS**
> - adrenal: **Cushing's**, tumours, *[added]* CAH
> - pituitary: **hyperprolactinaemia, acromegaly**
> - drugs: **anabolic steroids**
> - **Hypertension / HF (Reihan): β-blockers** induce or flare **psoriasis**.
> - **Tuberculosis (Kotb): erythema nodosum** (tender nodules; TB, sarcoid); **mycobacterial culture** of skin biopsies.
> - **Haematology 2: vitiligo** clusters with **pernicious anaemia** and autoimmune thyroid disease.
> - **FUO & Travel Medicine:** rash patterns as clues: **rose spots** (typhoid), dengue rash/tourniquet test.
> - **Psychiatry:** quality-of-life impact, measured with the **DLQI**.
> - **Psychotropic medication:** *[added]* lithium flares psoriasis and acne.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Analyse history and examination findings in skin disease	Exacerbating/relieving factors; drug triggers; occupational clues
Skill	⭐ Differentiate conditions by **morphology and distribution**; plan diagnostics	Wood's lamp colours; Auspitz/Nikolsky/dermatographism; biopsy type and transport medium

Main subtopics (slide 3):

- dermatological **history framework**
- systematic **skin examination**
- **investigations**: dermoscopy, biopsy, patch testing, **Wood's lamp**

> ⚠️ **Coverage gap: patch testing** is listed in the subtopics but never taught. *[added]* It is the test for **allergic (type IV) contact dermatitis**: allergens are applied under occlusion on the back and read at **48 h and 96 h**.

Framework (slide 4): History → Examination → Investigation → Management.

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2. Main Content (Lecture Order)

2.1 Dermatological History: Structure (Slide 5)

The same skeleton as any history, with skin-specific content:

1. Chief complaint
2. **History of presenting illness**
3. Past medical history
4. Systemic review
5. **Medications and allergies**
6. **Family history**
7. **Social and occupational history**
8. **Travel and environmental exposure**

2.2 History of Presenting Illness (Slides 6–12) ⭐⭐

Item	What to ask
Onset	When did the rash start? **First episode or recurrent?**
Time course of individual lesions	How long has each lesion been present? How long before it disappears? *[added: < 24 h → urticaria]*
Pattern of onset and spread	Where did it start, and how did it spread?
Site of first lesion	*[added: herald patch → pityriasis rosea]*
Symptoms	**Itchy? Painful? Burning? Tingling?** *[added: tingling before vesicles → HSV/VZV]*
Exacerbating / relieving factors	Below
Previous biopsy	Result
Previous treatment	Below
⭐ **Quality of life**	Effect on sleep, activities, social life, confidence, job; assessment tool: **Dermatology Life Quality Index (DLQI)**

Pigmented or suspicious lesions (slide 9): ask about evolution:

- **growth** or change in size?
- change in **shape/outline**?
- change in **colour**?
- **new symptoms**: **bleeding, pain, itch**?

[added] These map onto the **ABCDE** of melanoma: **A**symmetry, **B**order irregularity, **C**olour variation, **D**iameter > 6 mm, **E**volution. "E" is the most important and is exactly what slide 9 asks about.

Exacerbating factors (slide 10):

Factor	Example
UV exposure	Flare of **lupus** and **rosacea**
Temperature	**Hot drinks** flare **rosacea**
⭐ **Drugs**	**β-blocker-induced psoriasis**
Pregnancy	Flare of **psoriasis** (slide)

Factor	Example
Supplements	**Anabolic steroids → acne**

[added] Psoriasis often *improves* in pregnancy, though it can flare (especially pustular psoriasis of pregnancy) and commonly flares **postpartum**. The slide's example is narrower than reality.

Relieving factors (slide 11):

Factor	Example
UV exposure	**Sun improves psoriasis** (basis of **phototherapy**)
Moisturising	Helps **dry skin in dermatitis**
Drugs	**Antihistamines in urticaria**

> ⭐ **UV is both:** it **worsens lupus and rosacea** but **improves psoriasis**.

Previous treatment (slide 12):

- **steroid-responsive disease?** (supports an inflammatory dermatosis)
- **partial response → altered morphology**: ⭐ **tinea incognito** (topical steroid on a fungal infection removes the scale and active edge)
- **treatment-induced allergic/irritant contact dermatitis** (e.g. to a topical)
- ⭐ **hydroquinone-induced (exogenous) ochronosis**: paradoxical blue-black pigmentation after long-term bleaching cream
- **systemic treatment, phototherapy, laser** history

2.3 Past Medical History, Medications, Family & Social History (Slides 13–19) ⭐

Past medical history and systemic review (slide 13):

- **related comorbidities**
- **skin signs of systemic disease**, e.g. **PCOS** (acne, hirsutism, *[added]* acanthosis nigricans) and **diabetes** (*[added]* necrobiosis lipoidica, acanthosis, infections)
- ⭐ **contraindications to treatment**, e.g. **methotrexate in liver disease**

Medications and allergies (slide 14):

- contraindications
- drug interactions
- **drug-induced disease**, e.g. *[added]* SJS/TEN from allopurinol, carbamazepine, sulfonamides; fixed drug eruption

Family history (slide 15):

Purpose	Conditions
Genetic inheritance	**Atopic dermatitis, psoriasis, ichthyosis, melanoma**
Outbreak in family (contagious)	**Chickenpox, scabies, lice**

Social and occupational history (slides 16–17):

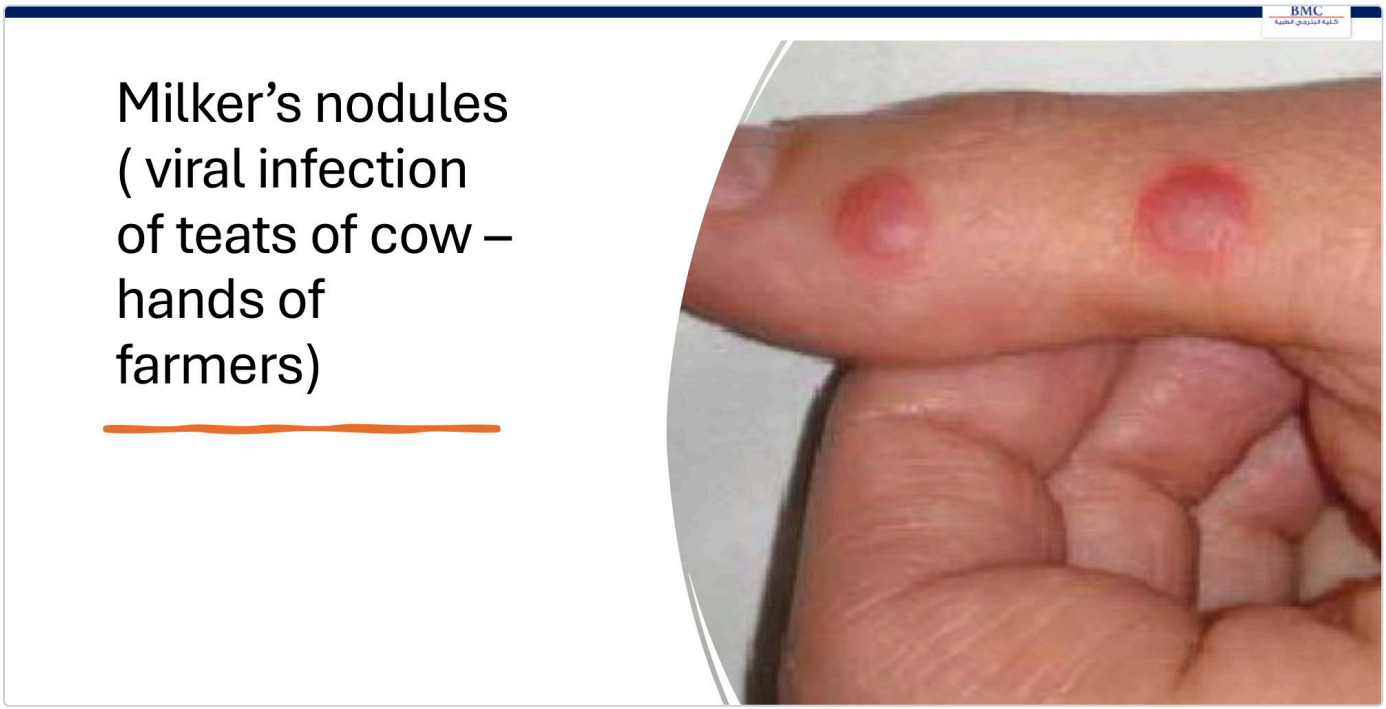
Factor	Effect
⭐ **Smoking**	Aggravates **lupus** and **hidradenitis suppurativa**

Factor	Effect
⭐ **Alcohol**	Worsens **rosacea**; **interacts with methotrexate** (hepatotoxicity)
Sun exposure	Sunburn, pigmentation, **wrinkles**, **skin cancer**
Sport	**Palmar calluses in weightlifters**; **athlete's foot in runners**
⭐ **Work-related contact dermatitis**	Irritant or allergic; look for a **time relation to exposure** and **improvement on vacation**
Work-related infection	⭐ **Milker's nodules**, ⭐ **orf**



Slide 18: Allergic contact dermatitis in hairdressers

[added] Hairdressers react to **para-phenylenediamine (PPD)** in hair dye and to glyceryl thioglycolate.



Slide 19: Milker's nodules

Milker's nodules:

- viral infection from **cows' teats**
- lesions on **farmers' hands**
- *[added]* paravaccinia, a **parapoxvirus**
- the clinically identical **orf** comes from **sheep/goats**
- both are self-limiting

> 🔁 **Recap: History.**
> - **Lesion timeline:** onset, recurrence, duration of each lesion, spread, first site, symptoms.
> - **Triggers:** UV (lupus/rosacea ↑, psoriasis ↓), **β-blockers → psoriasis**, anabolic steroids → acne, hot drinks → rosacea.
> - **Treatment history:** tinea incognito, hydroquinone ochronosis.
> - **Background:** methotrexate + liver disease/alcohol; smoking → lupus/HS; family history (atopy, psoriasis, melanoma; scabies outbreaks).
> - **Occupation:** vacation improvement = occupational dermatitis; milker's nodules/orf.
> - **Quality of life:** DLQI.
>
> 🔗 **Link:** history narrows the list; **morphology** clinches the diagnosis.
>
> ▶️ **Up next: Skin examination.**

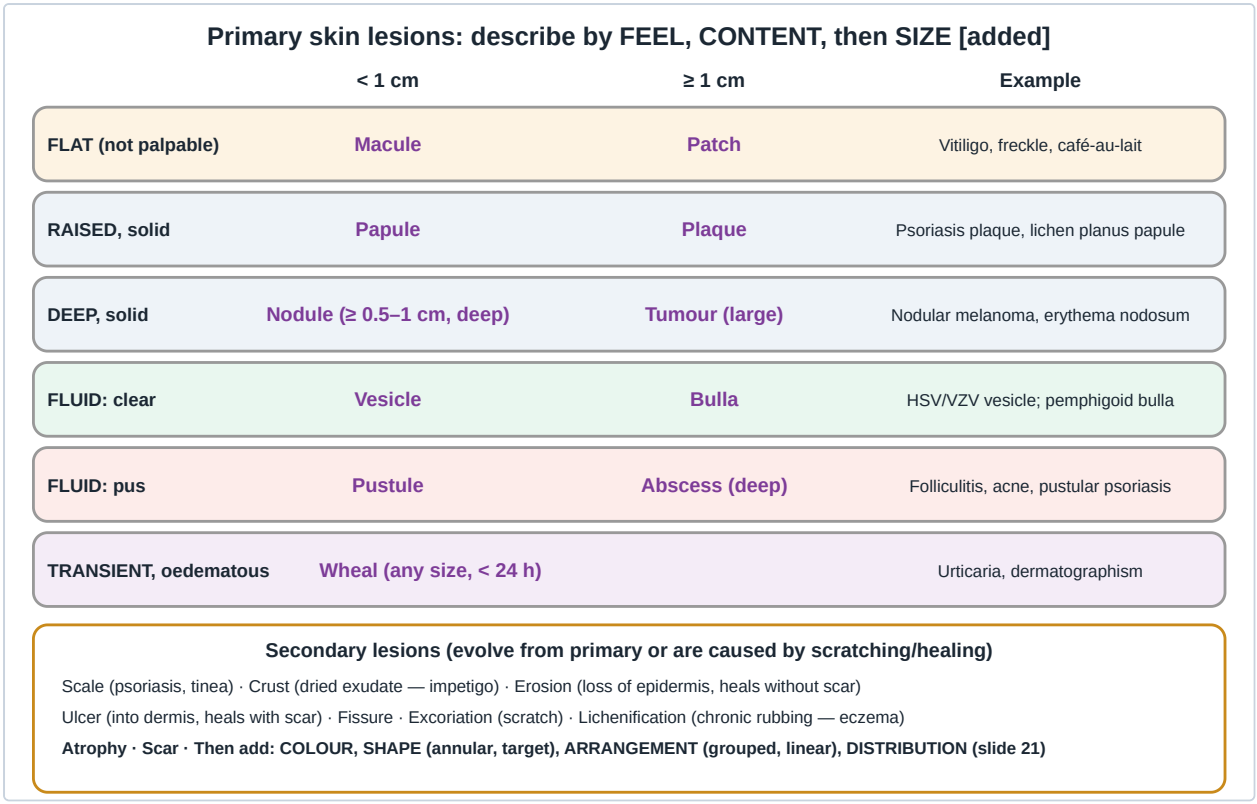
2.4 Skin Examination: Principles (Slides 20–22) ⭐⭐

Setting (slide 20):

- ⭐ **proper light**; **natural sunlight is best**
- **complete skin exam is ideal**, to detect **patterns or hidden lesions**
- include ⭐ **mucous membranes, scalp, hair and nails**
- **tools: magnifying lens, side lighting, dermatoscope, Wood's lamp**

Visual and tactile inspection: steps of morphology (slide 21):

1. ⭐ **Primary and secondary lesions**
2. **Colour**
3. **Shape**
4. **Configuration / arrangement**
5. **Distribution**



Added diagram: Primary and secondary skin lesions by feel, content and size

> *[added]* The lecture names "primary and secondary lesions" (and the summary slide lists macules, papules, plaques, nodules) but never defines them. The diagram fills that gap; the definitions are standard.

Palpation (slide 22):

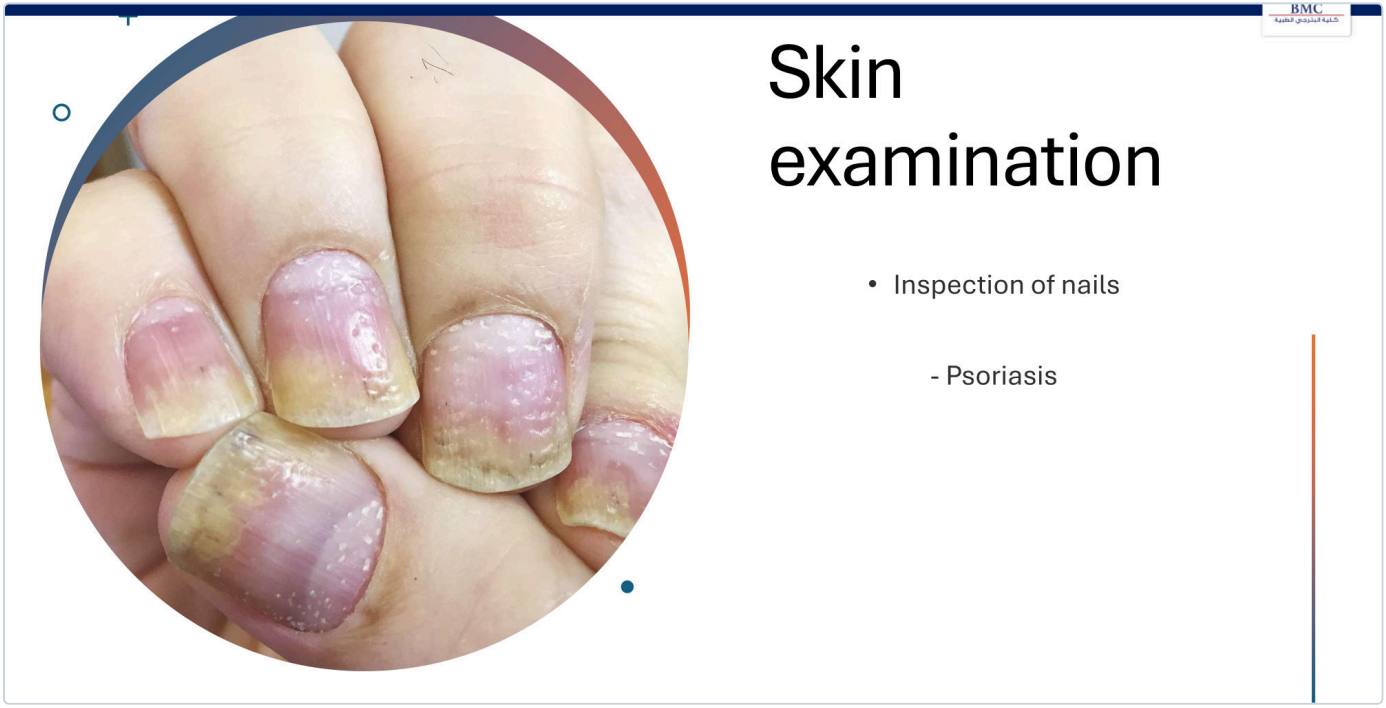
Property	Example condition
Temperature (warm)	**Cellulitis, abscess**
Tenderness	⭐ **Erythema nodosum**, **neuroma**
Consistency	**Nodular melanoma** (firm), **neurofibroma** (soft, "button-hole" sign *[added]*), **cyst**
Infiltration (firm, indurated)	**Cutaneous lymphoma, morphoea**
Texture	**Pityriasis versicolor** (fine scale on scratching; *[added]* "evoked scale")

2.5 Mucosa, Nails, Hair (Slides 23–26) ⭐



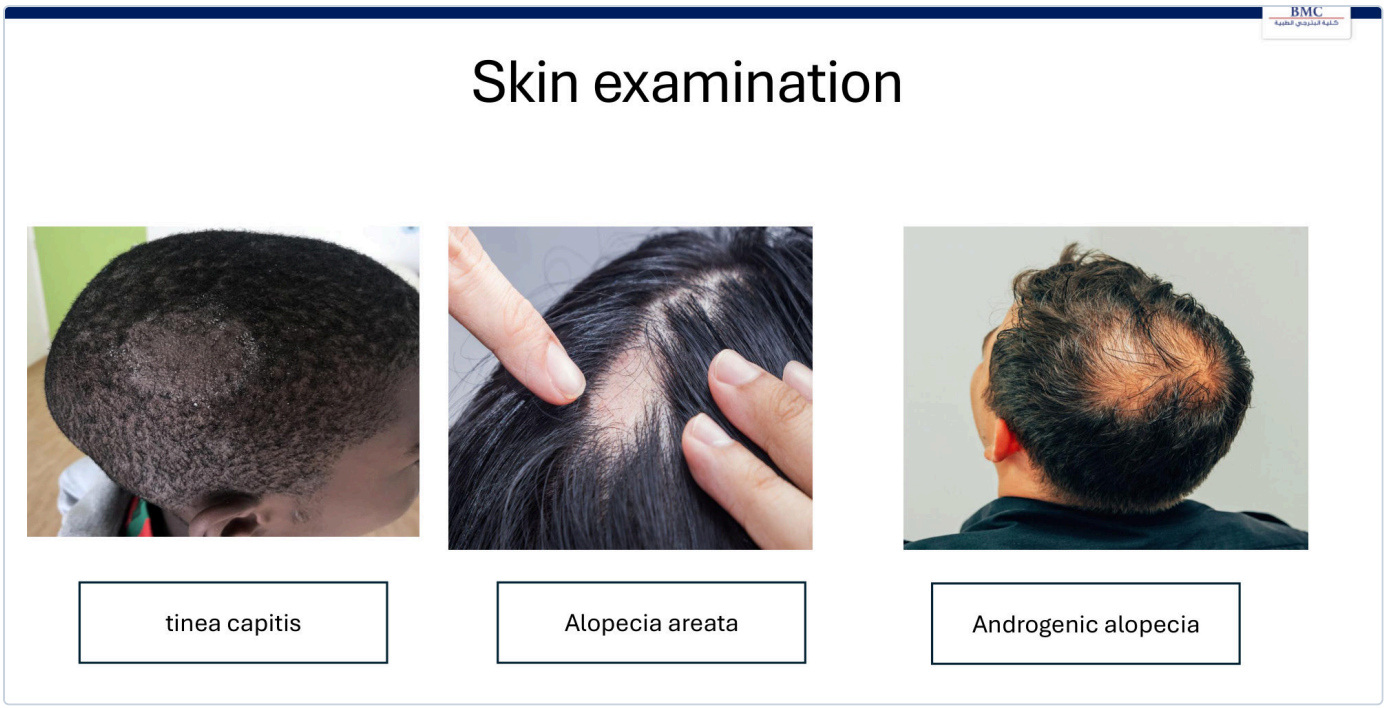
Slide 23: Oral lichen planus (white lacy Wickham striae) and cutaneous lesions

- **Mucosa: ⭐ lichen planus**. *[added]* White lacy **Wickham's striae** on the buccal mucosa; skin lesions are **purple, polygonal, pruritic, planar papules**. Associated with **hepatitis C** (slide 47).



Slide 24: Psoriatic nails (pitting, onycholysis, oil-drop discoloration)

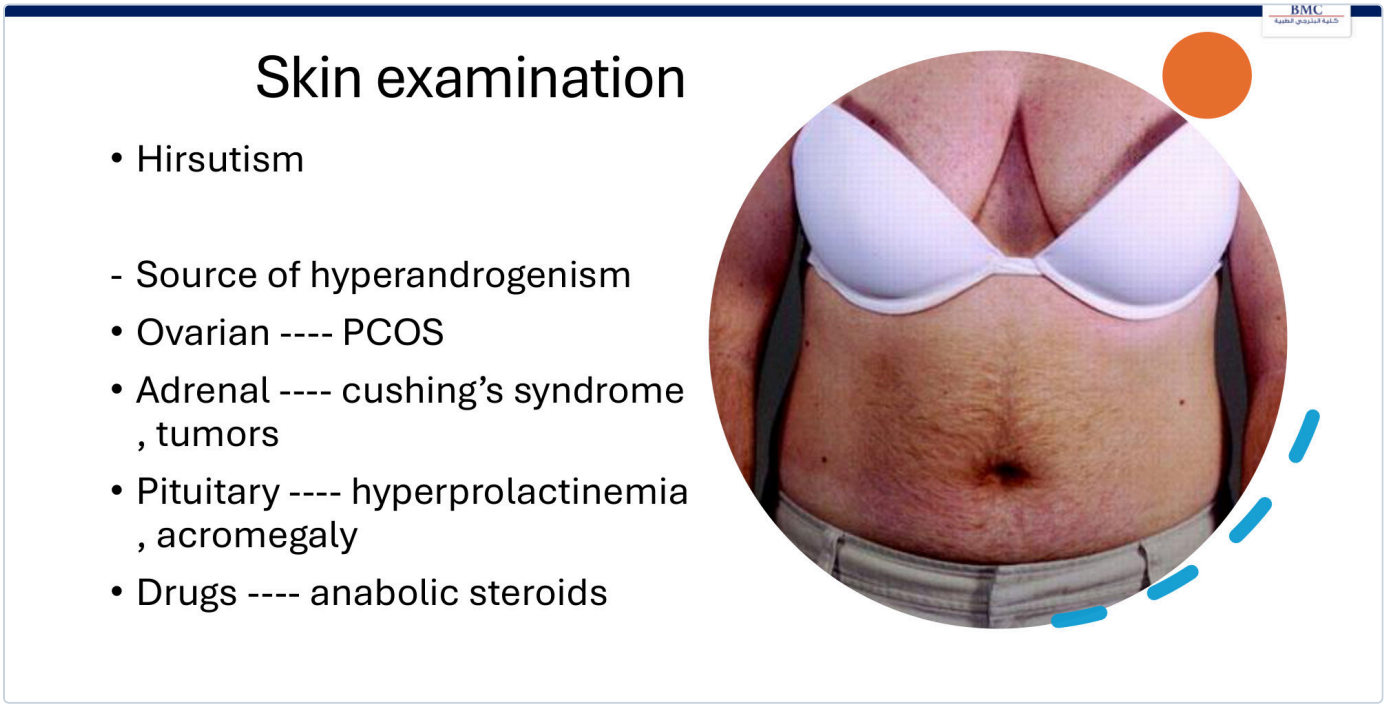
- **Nails: ⭐ psoriasis**. *[added]* Pitting, **onycholysis**, **oil-drop/salmon patch**, subungual hyperkeratosis; nail disease predicts **psoriatic arthritis**.



Slide 25: Alopecia: tinea capitis, alopecia areata, androgenetic alopecia

Hair: alopecia (slide 25):

Type	Clue ***[added]***
Tinea capitis	Scaly patch, broken hairs, ± lymphadenopathy, children; **Wood's lamp blue-green if *Microsporum***
Alopecia areata	Smooth round patch, **exclamation-mark hairs**; autoimmune (vitiligo, thyroid)
Androgenetic alopecia	Vertex/frontotemporal (men); diffuse crown thinning (women)

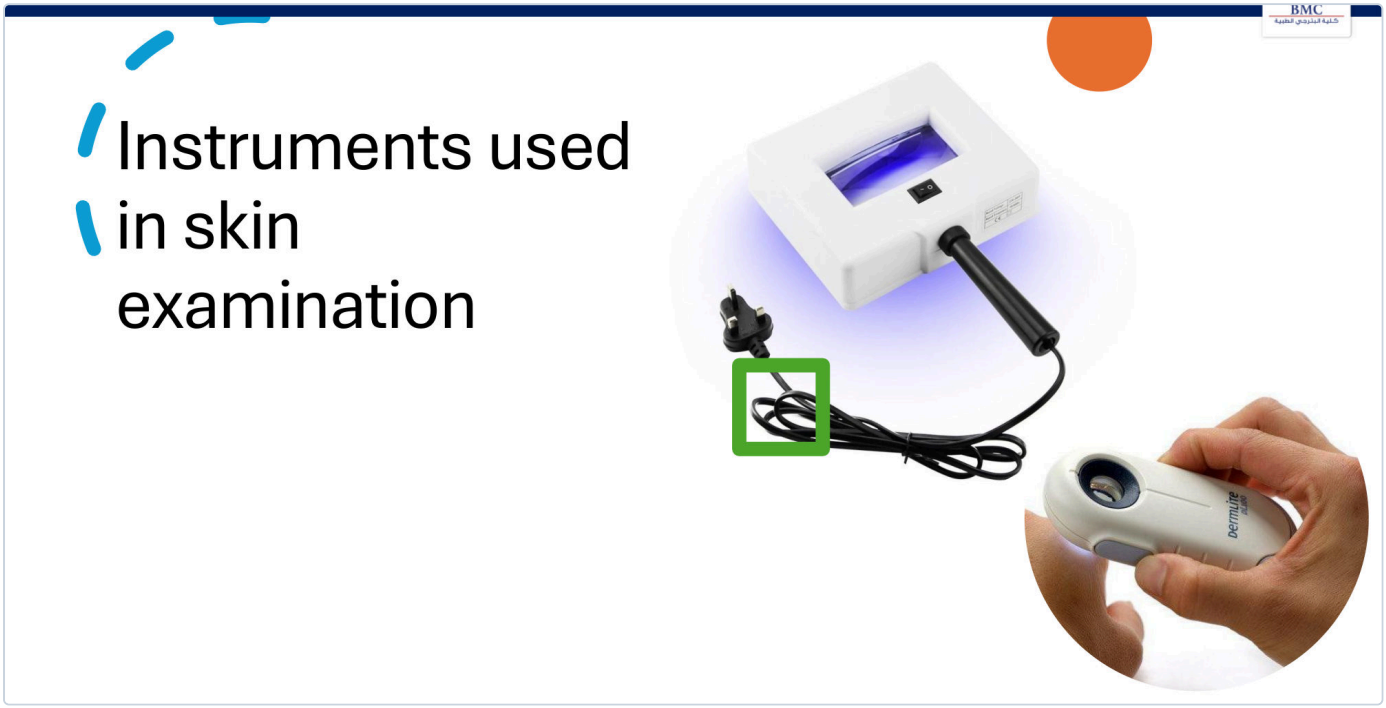


Slide 26: Hirsutism (male-pattern hair in a woman)

Hirsutism: source of hyperandrogenism (slide 26):

Source	Causes
Ovarian	⭐ **PCOS** (*[added]* ovarian tumour if rapid virilisation)
Adrenal	**Cushing's syndrome**, **tumours** (*[added]* **CAH**; DHEAS very high in adrenal carcinoma)
Pituitary	**Hyperprolactinaemia, acromegaly**
Drugs	**Anabolic steroids**

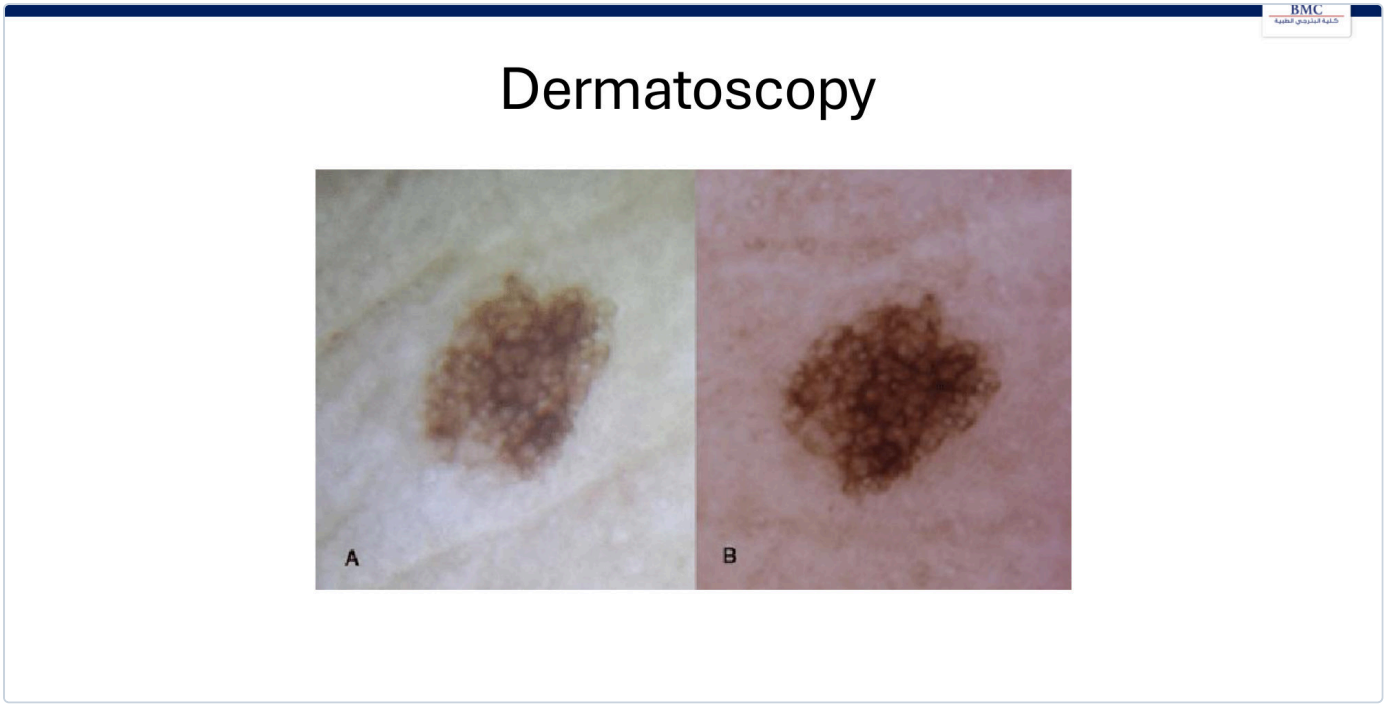
2.6 Instruments: Dermoscopy & Wood's Lamp (Slides 27–33) ⭐⭐⭐



Slide 27: Wood's lamp and hand-held dermatoscope

Dermatoscopy (slides 28–29):

- **hand-held magnifier** with (slide) **non-polarised light** (*[added]* polarised and contact non-polarised versions both exist)
- lesions and eruptions have **specific dermoscopic features**
- ⭐ main use: **atypical features of suspicious pre-cancerous or cancerous (pigmented) lesions**, *[added]* i.e. melanoma vs benign naevus



Slide 29: Dermoscopy of pigmented lesions

Wood's lamp (slides 30–31):

- **low-energy UV** (*[added]* UVA ~365 nm), examined in a **dark room**
- detects **fluorescent conditions** or the **borders** of some lesions
- ⭐ **normal skin does not fluoresce**

Colour	Condition	Organism / substance
⭐ **Coral red**	**Erythrasma**	***Corynebacterium minutissimum*** (porphyrins)
Yellow (*[added]* yellow-green / coppery-orange)	**Pityriasis versicolor**	***Malassezia furfur***
Bright (blue-)white	⭐ **Vitiligo** (depigmentation accentuated)	—
Blue-green	⭐ **Tinea capitis**	***Microsporum*** spp.
Pink / pink-red	**Porphyrins in urine** (*[added]* porphyria cutanea tarda)	—

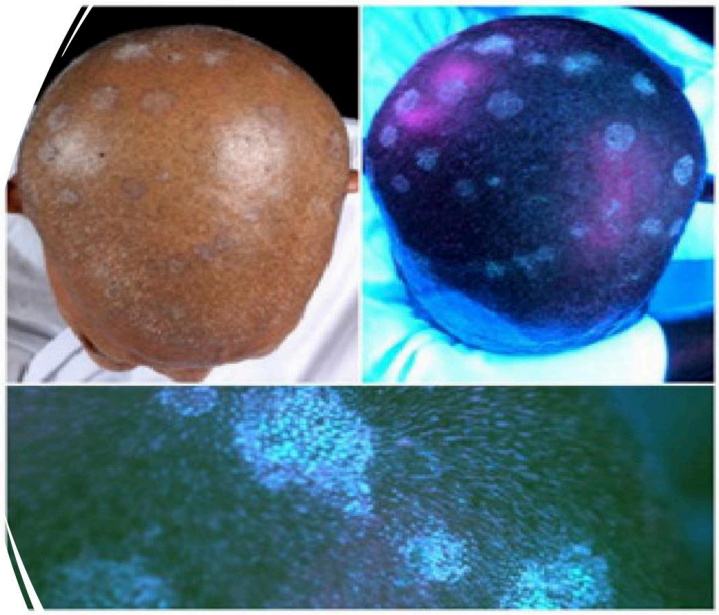
[added] **Other Wood's lamp uses:**

- **green**: *Pseudomonas* infection
- **vitiligo vs hypopigmentation**: vitiligo is depigmented → bright; pityriasis alba and postinflammatory hypopigmentation are less accentuated
- **melasma depth**: epidermal pigment is accentuated, dermal pigment is not

⚠️ *[added]* **Trap:** most tinea capitis in the Middle East/US is ***Trichophyton tonsurans*** (endothrix), which **does NOT fluoresce**. **A negative Wood's lamp does not exclude tinea capitis.** Send hair for KOH and culture.

Wood’s lamp

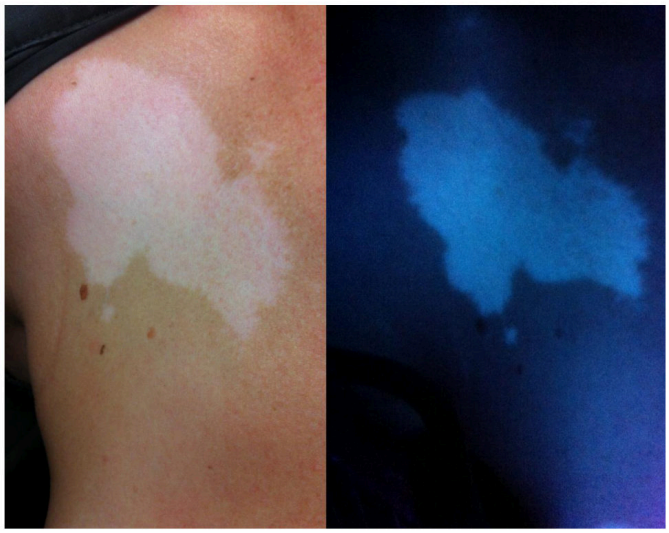
- Tinea capitis



Slide 32: Tinea capitis: blue-green fluorescence of affected hairs

Wood’s lamp

- Vitiligo



Slide 33: Vitiligo: bright white under Wood's lamp

🔁 **Recap: Examination & instruments.**

- **Setting:** daylight, whole skin + mucosa/scalp/hair/nails.
- **Morphology order:** primary/secondary → colour → shape → arrangement → distribution.
- **Palpation:** warm (cellulitis), tender (erythema nodosum), infiltrated (lymphoma, morphoea).
- **Dermoscopy:** suspicious pigmented lesions.
- **Wood's lamp colours: red erythrasma**, **yellow pityriasis versicolor**, **white vitiligo**, **blue-green *Microsporum***, **pink urine porphyrins**.

🔗 **Link:** some diagnoses come from **bedside manoeuvres** rather than instruments.

▶️ **Up next: Dermatological signs.**

2.7 Dermatological Signs (Slides 34–37) ⭐⭐⭐

Sign	How to elicit	Positive in
⭐ **Auspitz sign**	**Remove the scale → pinpoint bleeding**	⭐ **Psoriasis** (*[added]* dilated tortuous papillary capillaries under a thin suprapapillary epidermis)
⭐ **Nikolsky sign**	**Lateral finger pressure** on affected skin → **upper epidermis dislodges from lower layers**; on a blister → **lateral extension** of the blister (*[added]* = Asboe-Hansen / bulla-spread sign)	⭐ **TEN**, **SSSS**, **pemphigus vulgaris**
⭐ **Dermatographism**	**Stroke the skin** with a tongue depressor/spatula → **wheal and flare within minutes**	⭐ **Urticaria** (*[added]* symptomatic dermographism, the commonest physical urticaria)



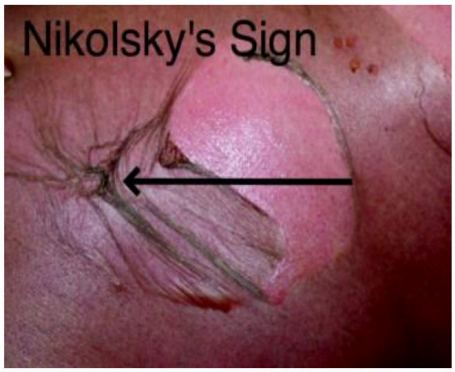
Slide 35: Auspitz sign (pinpoint bleeding after scale removal)

Nikolosky's sign

- Elicited by applying lateral pressure by finger in the affected skin , result in dislodgment of upper layers of epidermis from lower layers
- In case of blister , leads to lateral extension of blisters

- +ve in :

- Toxic epidermal necrolysis , staphylococcal scalded skin syndrome , pemphigus vulgaris



Slide 36: Nikolsky sign

Dermatographism

- Eliceted by stroking the skin using a tongue depressor or spatula . Within few minutes , a wheals and flare reaction develops

- +ve in urticaria



Slide 37: Dermatographism

[added] **Nikolsky: what's positive and why:**

Positive (split is intraepidermal or the epidermis is necrotic)	**Negative (split is subepidermal)**
Pemphigus vulgaris (anti-desmoglein 3 ± 1; suprabasal)	⭐ **Bullous pemphigoid** (anti-BP180/230; tense bullae)
SSSS (exfoliative toxin cleaves desmoglein 1; granular layer; **mucosa spared**)	Dermatitis herpetiformis
TEN/SJS (full-thickness necrosis; **mucosa involved**, drug trigger)	Bullous impetigo (localised)

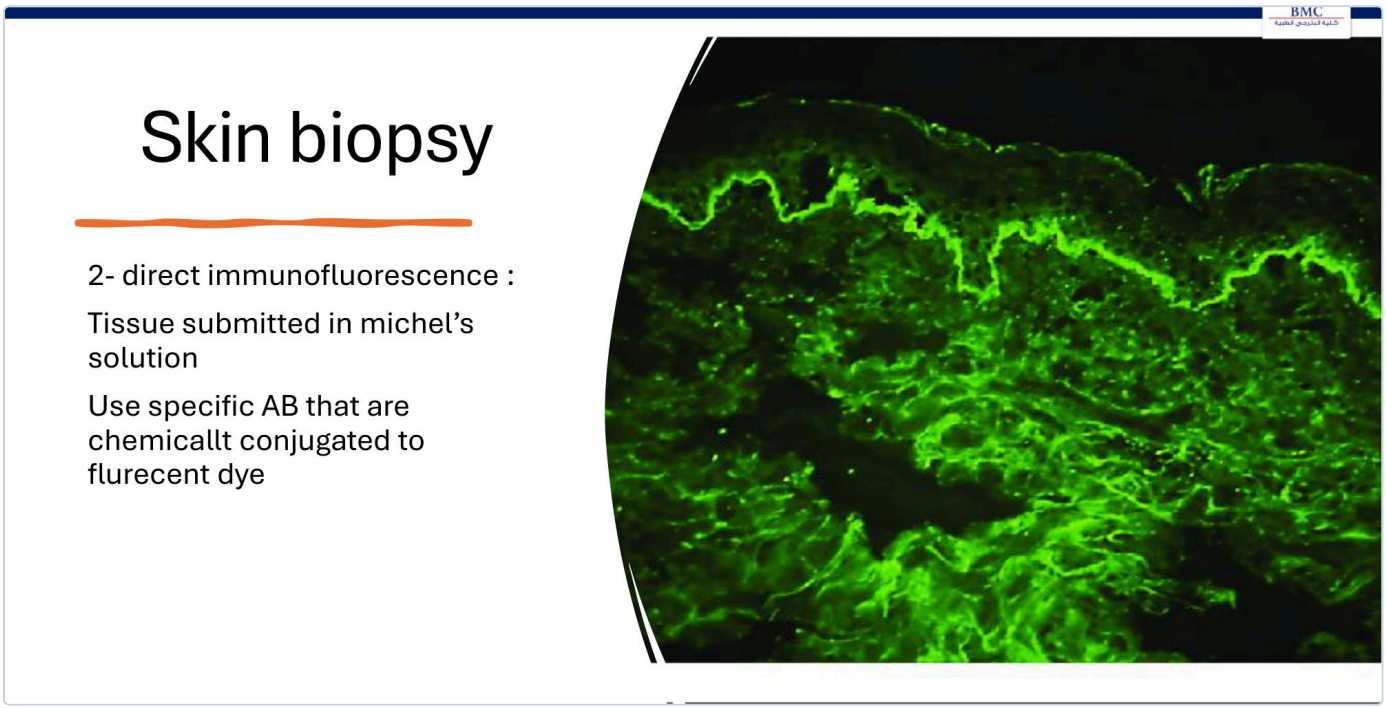
2.8 Investigations: Skin Biopsy (Slides 38–46) ⭐⭐

Indications (slide 40):

1. ⭐ **suspicious growth**: rule malignancy in or out
2. ⭐ **blistering disorders** (e.g. **bullous pemphigoid**)
3. **some inflammatory disorders**
4. **confirming tissue infection**

Three tests on skin specimens (slides 41–43):

Test	Medium	Stain / method	Use
1. Histopathology	⭐ **Formalin**	**H&E**	Most diagnoses, tumours
2. Direct immunofluorescence (DIF)	⭐ **Michel's solution** (*[added]* never formalin)	Antibodies conjugated to a **fluorescent dye**	⭐ **Blistering (autoimmune bullous) diseases**, vasculitis, lupus *[added]*
3. Tissue culture	⭐ **Normal saline** (sterile)	Culture	**Infection** (bacteria, mycobacteria, fungi)



Slide 42: Direct immunofluorescence

[added] **DIF patterns and site:**

- **pemphigus**: **intercellular "fishnet"** IgG/C3
- **bullous pemphigoid**: **linear** IgG/C3 at the **basement membrane**
- **dermatitis herpetiformis**: granular IgA in the dermal papillae
- biopsy **perilesional** skin for DIF, and the **lesion edge** for H&E

Types of skin biopsy (slides 44–46):

Type	Tool	Sample depth	Best for *[added]*
Shave	**Razor** blade	**Epidermis + part of dermis**	Superficial/raised lesions (seborrhoeic keratosis, BCC); **not** for suspected melanoma (cuts across depth)

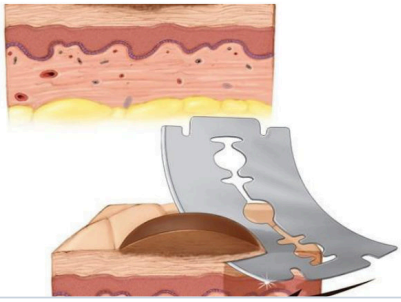
Type	Tool	Sample depth	Best for *[added]*
⭐ **Punch**	**Circular punch** (2–8 mm)	⭐ **All 3 layers: epidermis, dermis, superficial fat**	Inflammatory rashes, DIF, small lesions
⭐ **Excisional**	**Scalpel** (elliptical)	**Entire lump or a deep block**	⭐ **Suspected melanoma** (full depth for Breslow thickness, 1–3 mm margin), cysts, larger tumours

Types of skin biopsy

- Shave biopsy :

- Using razor (sample contain epidermis and portion of dermis)



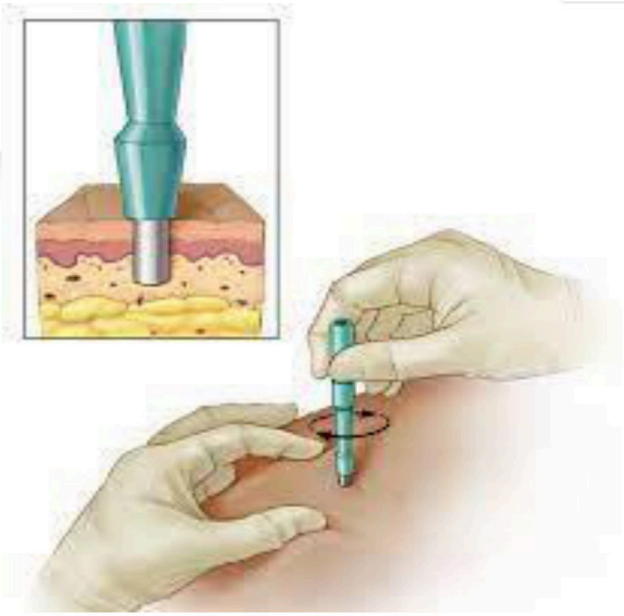


Slide 44: Shave biopsy

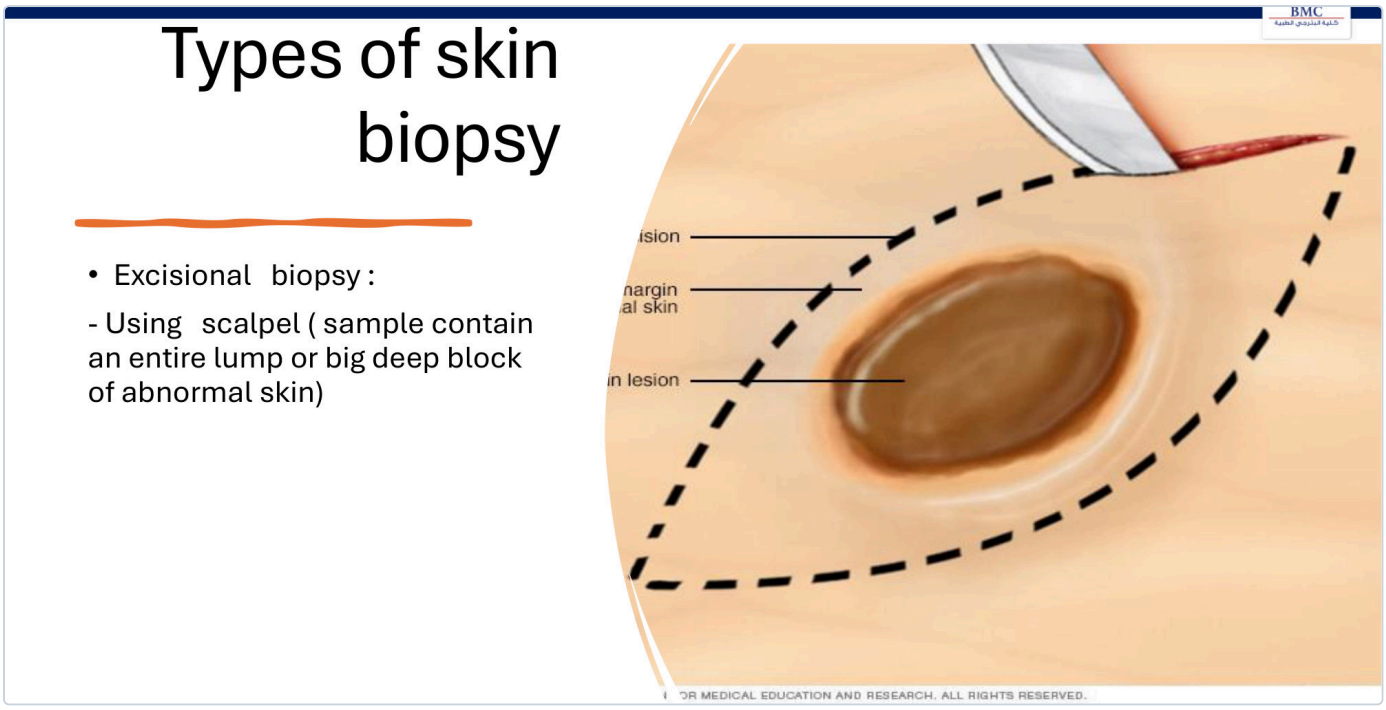
Types of skin biopsy

- Punch biopsy :

- Using circular punch tools (sample contain all 3 layers of skin : epidermis , dermis and superficial fat)



Slide 45: Punch biopsy



Slide 46: Excisional biopsy (elliptical, with margin)

2.9 Investigations: Blood & Microbiology (Slides 47–51) ⭐

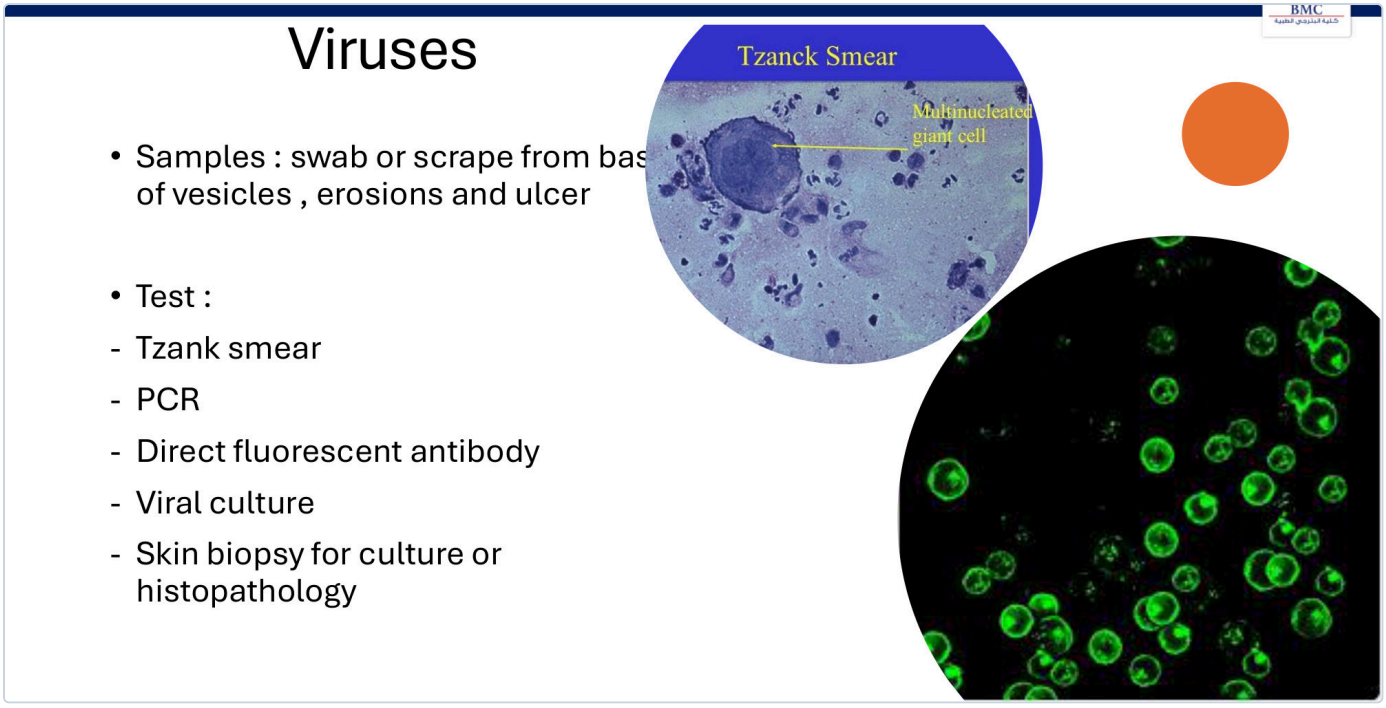
Blood work-up (slide 47):

Purpose	Example
Screen for an underlying systemic cause	**Itch** (*[added]* FBC, ferritin, renal/liver/thyroid tests, glucose), **hair loss** (*[added]* ferritin, TSH)
Screen for associated systemic disease	⭐ **Lichen planus → hepatitis C**
Hormonal studies	**Acne/hirsutism → PCOS** (*[added]* testosterone, DHEAS, LH/FSH, prolactin)
⭐ **Drug monitoring**	**Isotretinoin** (*[added]* lipids, LFTs, pregnancy test), **methotrexate** (*[added]* FBC, LFTs, renal)

Microbiological tests (slide 48):

- **swab** for microscopy, culture and sensitivity
- **swab/scraping of a vesicle base** for viral tests
- **skin scraping, plucked hair, nail** for fungal tests
- ⭐ **bedside direct microscopy** for **scabies and fungi**

Organism	Samples	Tests
Fungi (slide 49)	**Skin scraping, nail clipping, plucked hair**	⭐ **KOH microscopy** (hyphae); **fungal culture**; **biopsy with fungal stain** (*[added]* PAS)
Bacteria & mycobacteria (slide 50)	Swab from **vesicles, pustules, erosions, ulcers**	**Staining and microscopy** (Gram, *[added]* ZN); **bacterial/mycobacterial culture**; **biopsy for tissue culture**
Viruses (slide 51)	Swab/scrape from the **base** of vesicles, erosions, ulcers	⭐ **Tzanck smear** (*[added]* multinucleated giant cells; **HSV and VZV**; can't tell them apart); ⭐ **PCR** (*[added]* test of choice); **direct fluorescent antibody**; viral culture; biopsy



Slide 51: Tzanck smear (multinucleated giant cells) and DFA

🔁 **Recap: Investigations.**

- **Biopsy indications:** suspected cancer, **blistering disease**, inflammation, infection.
- **Transport medium: formalin (H&E)**, **Michel's (DIF)**, **saline (culture)**.
- **Biopsy type: shave** = epidermis + part of dermis; **punch** = all 3 layers; **excision** = whole lesion.
- **Blood tests:** systemic cause, **lichen planus → HCV**, PCOS hormones, isotretinoin/MTX monitoring.
- **Microbiology: KOH** for fungi; Gram/culture for bacteria; **Tzanck/PCR** for vesicles.

2.10 Summary & Case (Slides 52–54)

Summary slide:

- detailed history (onset, progression, triggers, drugs)
- systematic exam with **morphology** (macules, papules, plaques, nodules)
- **targeted tools**: **dermoscopy** at the bedside, **biopsy** as the definitive test

Case scenario

- 42-year-old obese man with type 2 diabetes presents with a 3-month history of a well-demarcated, reddish-brown, slightly scaly patch in his left inguinal fold.
- Bedside Diagnostic Tool: Examination under Wood's Lamp in a darkened room exhibits a striking bright coral-red fluorescence.



Slide 53: Case: coral-red fluorescence in the groin

See Q1 in §7.

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3. High-Yield Comparison Tables

Table A: Wood's Lamp Colours

Colour	Diagnosis	Cause
Coral red	**Erythrasma**	*C. minutissimum* (porphyrin)
Yellow / yellow-green	**Pityriasis versicolor**	*Malassezia furfur*
Bright blue-white	**Vitiligo**	Absent melanin
Blue-green	**Tinea capitis**	***Microsporum*** (not *T. tonsurans*) *[added]*
Pink-red (urine)	**Porphyria**	Uroporphyrins *[added: PCT]*
Green *[added]*	*Pseudomonas*	Pyoverdin
None	**Normal skin**	—

Table B: Bedside Signs

Sign	Elicitation	Disease
Auspitz	Scrape scale → **pinpoint bleeding**	**Psoriasis**
Nikolsky	Lateral pressure → **epidermis shears / blister spreads**	**Pemphigus vulgaris, SSSS, TEN** (*[added]* **negative in bullous pemphigoid**)
Dermatographism	Stroke → **wheal and flare** in minutes	**Urticaria**
[added] **Koebner**	Trauma → new lesions along the trauma line	**Psoriasis, lichen planus, vitiligo** (and warts, molluscum: pseudo-Koebner)
[added] **Darier**	Rub lesion → urticates	**Mastocytosis** (urticaria pigmentosa)

Table C: Skin Biopsy

	Shave	Punch	Excisional
Tool	Razor	Circular punch	Scalpel
Depth	Epidermis + part of dermis	**Epidermis + dermis + fat**	Whole lesion
Use *[added]*	Superficial lesions	**Rashes, DIF**	**Suspected melanoma**

Table D: Specimen Transport

Test	Medium
H&E histology	**Formalin**
DIF	**Michel's solution**
Culture	**Normal saline**

Table E: Exacerbating / Relieving Factors

Factor	Worsens	Improves
UV/sun	**Lupus, rosacea** (*[added]* porphyria, polymorphic light eruption)	**Psoriasis** (*[added]* atopic dermatitis often)
Drugs	**β-blockers → psoriasis**; **anabolic steroids → acne** (*[added]* lithium → psoriasis/acne)	**Antihistamines → urticaria**
Smoking	**Lupus, hidradenitis suppurativa** (*[added]* palmoplantar pustulosis)	—
Alcohol	**Rosacea**; **MTX interaction**	—
Heat / hot drinks	**Rosacea**	—
Moisturisers	—	**Dry skin/dermatitis**

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
UV is harmful to skin	**UV improves psoriasis** (phototherapy) but **worsens lupus and rosacea**
Steroid response confirms inflammation	**Tinea incognito**: steroids partly suppress a fungal infection → atypical morphology; **KOH before steroids**
Bleaching creams lighten	**Hydroquinone → exogenous ochronosis** (darkening)
Tinea capitis fluoresces blue-green	*[added]* Only ***Microsporum***; ***T. tonsurans*** (commonest) **doesn't** → negative Wood's lamp ≠ no tinea
Normal skin under Wood's lamp	**Doesn't fluoresce**
Nikolsky positive = pemphigus	Also **SSSS and TEN**; *[added]* **negative in bullous pemphigoid**
Biopsy in formalin	**DIF needs Michel's**; culture needs **saline**
Shave biopsy for pigmented lesions	*[added]* ❌ For suspected melanoma → **excisional** (full thickness for Breslow depth)
Occupational dermatitis	**Improves on vacation**; look for a time relation to exposure
Psoriasis "flares in pregnancy" (slide)	*[added]* Often **improves** in pregnancy and flares **postpartum**

Pattern recognition:

- **obese diabetic + reddish-brown groin patch + coral-red fluorescence → erythrasma**
- **scaly plaques + nail pitting + pinpoint bleeding on scraping → psoriasis**
- **flaccid blisters + oral erosions + Nikolsky + → pemphigus vulgaris**
- **tense blisters in an elderly patient + Nikolsky − → bullous pemphigoid** *[added]*
- **purple polygonal papules + lacy oral striae → lichen planus → check HCV**
- **farmer + nodules on fingers after milking → milker's nodules**

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5. Clinical Correlations

- **Prescribing a β-blocker** to a patient with psoriasis may flare it. Ask about skin disease when starting antihypertensives (link to the HTN lecture).
- **Methotrexate for psoriasis:**
  - check **liver disease and alcohol** (contraindication and interaction)
  - monitor **FBC/LFTs**
  - *[added]* give folic acid (link to Haematology 2: methotrexate → folate deficiency)
- **Isotretinoin:**
  - monitor **lipids and LFTs**
  - *[added]* **pregnancy prevention** (teratogen)
- **Hirsutism** → endocrine work-up (PCOS, Cushing's, CAH, prolactin, acromegaly). Rapid virilisation → tumour.
- **Lichen planus** → **hepatitis C** serology.
- **Before topical steroids on a scaly rash** → **KOH scraping** to avoid tinea incognito.
- **Blistering disease** → **two biopsies**: lesional (formalin, H&E) + perilesional (**Michel's, DIF**) *[added]*.
- **Suspicious pigmented lesion** → history of **evolution** (ABCDE) + **dermoscopy** → **excisional biopsy**.

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6. Rapid Revision (Last-Day)

- **Approach:** history → exam → investigation → management.
- **History sections:** CC, HPI, PMH, systemic review, **drugs/allergies**, **family**, **social/occupational**, **travel/environment**.
- **HPI:** onset, first vs recurrent, **duration of individual lesions**, onset/spread pattern, **first site**, symptoms (itch, pain, burning, tingling), triggers, previous biopsy/treatment, **QoL (DLQI)**.
- **Suspicious pigmented lesion: growth, shape, colour change, bleeding/pain/itch** (evolution).
- **Exacerbating factors:**
  - **UV → lupus, rosacea**
  - **hot drinks → rosacea**
  - **β-blocker → psoriasis**
  - **pregnancy → psoriasis** (slide)
  - **anabolic steroids → acne**
- **Relieving factors: sun → psoriasis**; **moisturiser → dermatitis**; **antihistamine → urticaria**.
- **Previous treatment: tinea incognito**; treatment-induced contact dermatitis; **hydroquinone ochronosis**.
- **Background history:**

    - PMH: PCOS, DM; **methotrexate contraindicated in liver disease**
    - family history: atopic dermatitis, psoriasis, ichthyosis, melanoma; outbreaks of chickenpox, scabies, lice
- **Social and occupational history:**
    - **smoking → lupus, hidradenitis**
    - **alcohol → rosacea, MTX interaction**
    - sun → burn, pigment, wrinkles, cancer
    - sport → **weightlifters' palmar calluses, runners' athlete's foot**
    - **work contact dermatitis improves on vacation**
    - **milker's nodules (cow teats), orf**
    - hairdressers → allergic contact dermatitis
- **Exam setting: natural daylight**, complete exam incl. **mucosa, scalp, hair, nails**; tools: lens, side light, dermatoscope, Wood's lamp.
- **Morphology order: primary/secondary → colour → shape → arrangement → distribution**.
- **Palpation clues:**
    - **warm**: cellulitis/abscess
    - **tender**: erythema nodosum, neuroma
    - **consistency**: nodular melanoma, neurofibroma, cyst
    - **infiltrated**: cutaneous lymphoma, morphoea
    - **texture**: pityriasis versicolor
- **Mucosa, nails, hair:**
    - **mucosa: lichen planus**; **nails: psoriasis**
    - **alopecia**: tinea capitis, areata, androgenetic
    - **hirsutism**: ovarian (PCOS), adrenal (Cushing's, tumour), pituitary (prolactin, acromegaly), drugs (anabolic)
- **Dermoscopy:** hand-held, **suspicious pigmented / pre-cancerous lesions**.
- **Wood's lamp:**
    - low-energy UV; **normal skin doesn't fluoresce**
    - **red = erythrasma (*C. minutissimum*)**
    - **yellow = pityriasis versicolor (*M. furfur*)**
    - **bright white = vitiligo**
    - **blue-green = tinea capitis (*Microsporum*)**
    - **pink = urine porphyrins**
- **Bedside signs:**
    - **Auspitz** (scale off → pinpoint bleeding) = **psoriasis**
    - **Nikolsky** (lateral pressure → epidermis separates / blister spreads) = **TEN, SSSS, pemphigus vulgaris**
    - **dermatographism** (stroke → wheal and flare) = **urticaria**
- **Biopsy indications:** malignancy, **blistering (pemphigoid)**, inflammation, infection.
- **Biopsy specimens: formalin/H&E**; **DIF in Michel's**; **culture in saline**.
- **Biopsy types: shave** (razor; epidermis + part of dermis), **punch** (all 3 layers incl. fat), **excisional** (scalpel; whole lesion).
- **Blood tests:** systemic cause of itch/hair loss; **LP → HCV**; **PCOS hormones**; **isotretinoin/MTX monitoring**.
- **Microbiology:**
    - **fungi**: scraping/nail/hair → **KOH**, culture, biopsy
    - **bacteria/mycobacteria**: swab → stain, culture, tissue culture
    - **viruses**: vesicle-base swab → **Tzanck, PCR, DFA**, culture, biopsy
    - **bedside microscopy for scabies and fungus**

7. Exam-Style Questions

Q1. (Lecture case) A 42-year-old obese man with type 2 diabetes has a 3-month, well-demarcated, **reddish-brown, slightly scaly patch** in the **left inguinal fold**. Wood's lamp in a dark room shows **bright coral-red fluorescence**. Diagnosis and cause of the fluorescence? A. Tinea cruris (*T. rubrum*) · **B. Erythrasma (*Corynebacterium minutissimum*)** · C. Intertriginous candidiasis · D. Inverse psoriasis

> **Answer: B** *[reasoning added; key not on slides]*.
>
> - **Coral-red fluorescence = erythrasma** (slide 31): *C. minutissimum* produces **porphyrins** (coproporphyrin III).
> - Obesity and diabetes are classic risk factors (warm, moist flexures).
> - **Why not the others:**
> - **tinea cruris** has an **active scaly advancing edge with central clearing** and **doesn't fluoresce** (*Trichophyton*)
> - **candidiasis**: beefy-red with **satellite pustules**, no coral fluorescence
> - **inverse psoriasis**: shiny red, non-scaly, no fluorescence
> - *[added]* **Treatment:** topical **erythromycin/clindamycin** or fusidic acid; oral erythromycin/clarithromycin if extensive.

Q2. A 55-year-old has flaccid blisters and painful **oral erosions**. Lateral pressure on normal-looking skin next to a blister shears off the epidermis. Which sign is this, and which biopsy medium confirms the diagnosis by immunofluorescence? A. Auspitz sign; formalin · **B. Nikolsky sign; Michel's solution** · C. Dermatographism; saline · D. Koebner phenomenon; formalin · E. Darier sign; Michel's

> **Answer: B** (slides 36, 42).
>
> - Nikolsky positive → **pemphigus vulgaris** (also TEN, SSSS).
> - **DIF** specimen goes in **Michel's**. *[added]* Expect an intercellular IgG "fishnet" pattern.

Q3. A patient with silvery scaly plaques on the elbows has pinpoint bleeding when the scale is scraped off. The rash began after a new antihypertensive. Most likely culprit drug? A. Amlodipine · **B. Propranolol** · C. Hydrochlorothiazide · D. Lisinopril · E. Doxazosin

> **Answer: B** (slides 10, 35).
>
> - **Auspitz sign = psoriasis**; **β-blockers** induce or flare it.
> - *[added]* Lithium, antimalarials and abrupt steroid withdrawal also flare psoriasis.

Q4. A pigmented lesion on the back has grown and changed colour over 6 months and now bleeds occasionally. Dermoscopy shows atypical features. Best biopsy? A. Shave · B. 2-mm punch from the edge · **C. Excisional biopsy with narrow margin** · D. KOH scraping · E. Tzanck smear

> **Answer: C** *[slides 9, 28, 46 + added]*.
>
> - Suspected **melanoma** → excise the **whole lesion** so **Breslow depth** can be measured.
> - A shave biopsy may transect the lesion and under-stage it.

Q5. A child has a scaly patch of hair loss with broken hairs. Wood's lamp shows **no fluorescence**. Best next step? A. Diagnose alopecia areata; topical steroid · **B. Plucked hairs and scalp scrapings for KOH and fungal culture** · C. Reassure: tinea capitis excluded · D. Punch biopsy for DIF · E. Hormonal studies

> **Answer: B** *[slides 25, 31, 49 + added]*.
> - Only ***Microsporum*** fluoresces blue-green. ***T. tonsurans***, the commonest cause, doesn't.
> - A negative Wood's lamp does not exclude tinea; confirm with KOH and culture.

Q6. A dairy farmer has two firm, red-violet nodules on his fingers 1 week after milking cows. Diagnosis? A. Orf · **B. Milker's nodules** · C. Sporotrichosis · D. Cutaneous anthrax · E. Pyogenic granuloma

> **Answer: B** (slides 17, 19).
> - **Cows' teats → milker's nodules** (parapoxvirus). **Orf** comes from **sheep/goats**.
> - *[added]* Self-limiting in 4–8 weeks.

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8. Exam Pitfalls & Traps

Trap	Correct thinking
Any red groin rash = tinea cruris	❌ **Coral-red fluorescence = erythrasma** (bacterial; *C. minutissimum*)
Negative Wood's lamp excludes tinea capitis	❌ *[added]* ***T. tonsurans*** doesn't fluoresce; do KOH and culture
Normal skin fluoresces faintly → abnormal	❌ **Normal skin doesn't fluoresce** (slide)
Nikolsky positive in all blistering disease	❌ **TEN, SSSS, pemphigus vulgaris**; *[added]* **negative in bullous pemphigoid**
Auspitz sign = lichen planus	❌ **Psoriasis**
Dermatographism = vasculitis	❌ **Urticaria**
Put every skin biopsy in formalin	❌ **DIF → Michel's**; **culture → saline**
Punch biopsy reaches only the dermis	❌ **All 3 layers incl. superficial fat**
Shave biopsy for suspected melanoma	❌ *[added]* **Excisional** (full thickness)
Sun always worsens skin disease	❌ **Improves psoriasis**; worsens lupus and rosacea
Steroid cream for every itchy scaly rash	❌ **Tinea incognito**; scrape for KOH first
Hydroquinone always lightens	❌ Long-term use → **exogenous ochronosis**
Milker's nodules = orf	❌ Same virus family; **cows** (milker's) vs **sheep/goats** (orf)
Patch testing taught in this lecture	⚠️ Listed in the subtopics but **not taught**; *[added]* for **allergic contact dermatitis**, read at 48/96 h
Lichen planus is purely a skin disease	❌ Screen for **hepatitis C** (slide 47)